Exploratory analysis of timing and resource distributions

Jiachen Cong

University of Illinois Urbana-Champaign
Department of Statistics

Mar 11, 2026

Contents

1. Waiting time distribution
2. Submission patterns
3. Runtime distribution for completed jobs
4. Analysis for requested resource
5. Is there a significant difference in the waiting time between tasks that use only the CPU and those that require a GPU?
6. Plan for future work

Waiting time distribution

Waiting time distribution

- I use the terminal table ("master_2025_joblevel_terminal.csv.gz"), since it has Start/End and WaitTimeSec.
- Not included: tasks in PENDING status; some tasks in CANCELLED status (tasks cancelled and not started)

Waiting time distribution

Statistic	Wait Time (hours)
Observations	5,517,282
Mean	6.95
Std. Dev.	31.27
Min	0.00
Median	0.42
90th Percentile	13.01
95th Percentile	31.13
99th Percentile	116.10
Max	6865.70

Table 1: Wait time summary

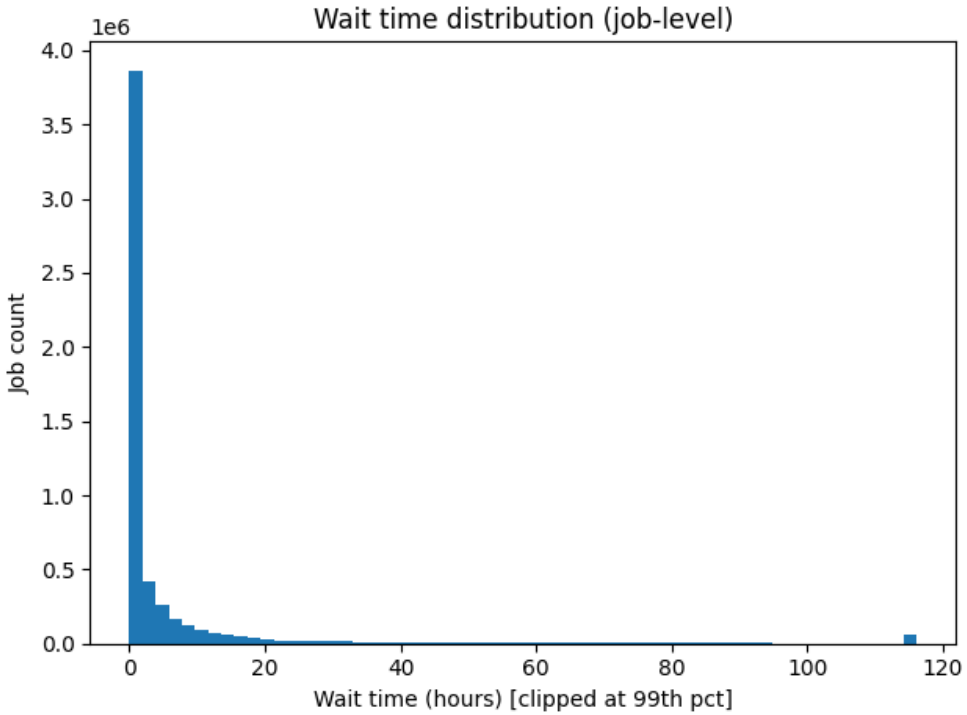

Figure 1: Wait time distribution (job-level)

Submission patterns

Job submissions by weekday

Weekday	Number of Submissions
Monday	864,472
Tuesday	969,666
Wednesday	837,331
Thursday	814,956
Friday	855,459
Saturday	816,110
Sunday	867,332

Table 2: Job submissions by weekday

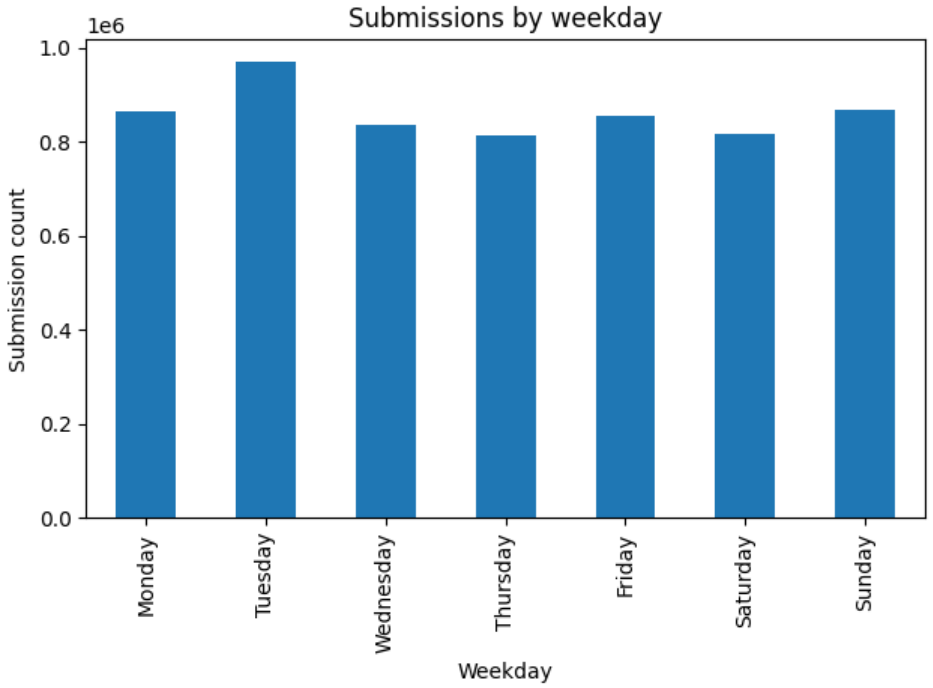

Figure 2: Submissions by weekday

Note: In this section, I use the data set "master_2025_joblevel_submission.csv.gz".

Job submissions by hour of day

Hour	Submissions	Hour	Submissions
0	271,256	12	295,872
1	240,464	13	279,835
2	220,156	14	304,868
3	139,365	15	377,099
4	123,713	16	359,670
5	127,304	17	322,188
6	137,256	18	294,654
7	143,170	19	256,135
8	179,919	20	244,337
9	190,417	21	282,626
10	276,210	22	350,138
11	300,606	23	308,068

Table 3: Job submissions by hour of day

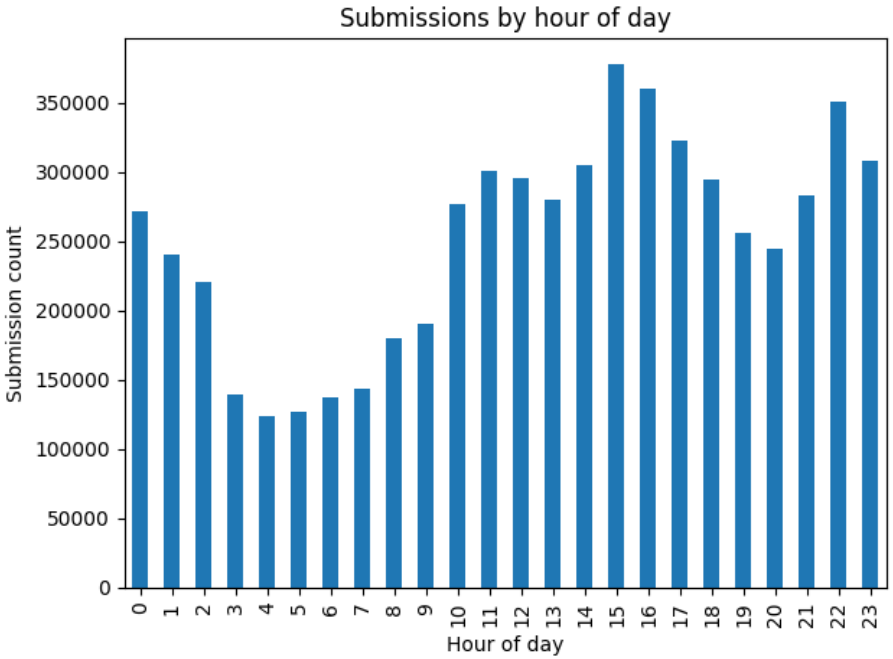

Figure 3: Submissions by hour of day

Note: In this section, I use the data set "master_2025_joblevel_submission.csv.gz".

Runtime distribution for completed jobs

Some instructions

- In this section, I only summarize the running time for COMPLETED tasks. Since only these finished tasks have recorded start and end time.
- I removed negative values.
- Negative runtime count: 20 / 4,640,994 (0.000431%).
- In this section, I used the dataset "master_2025_joblevel_terminal.csv.gz".

Runtime summary

Statistic	Runtime (hours)
Observations	4,640,974
Mean	0.559
Std. Dev.	3.363
Min	0.000
Median (50%)	0.019
90th Percentile	1.059
95th Percentile	1.946
99th Percentile	10.398
Max	627.116

Table 4: Runtime summary

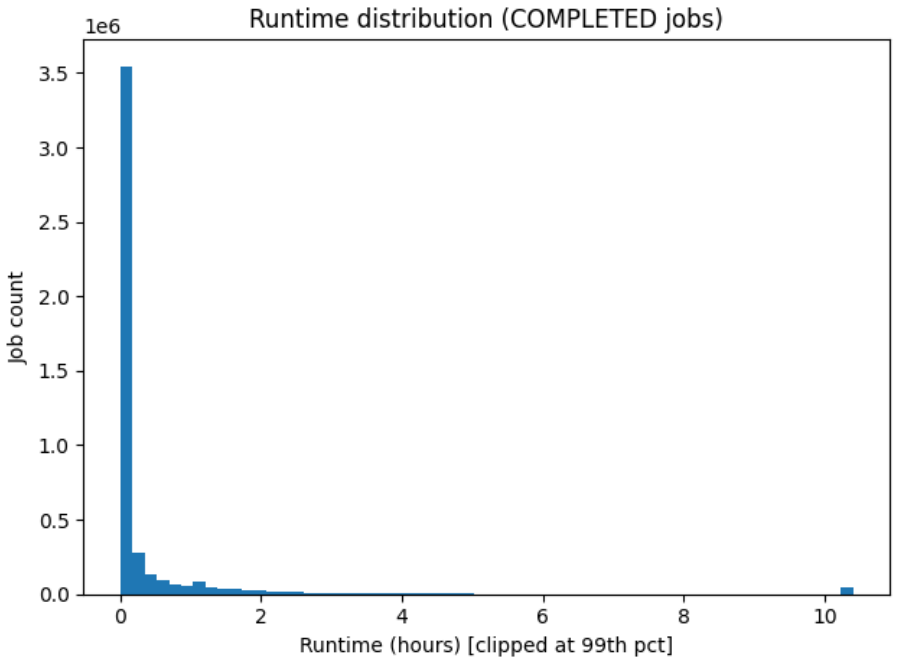

Figure 4: Runtime distribution

Analysis for requested resource

Instructions

- In this section, I used the dataset "master_2025_joblevel_submission.csv.gz" in the analysis of Required CPUs, Required Nodes, and Required Memory; while in the analysis of Required GPUs, I used the dataset "master_2025_joblevel_terminal.csv.gz".
- I extracted the data of GPUs' usage from the column 'ReqTRES'. For jobs that require GPUs, the corresponding ReqTRES column contains entries such as "gres/gpu" . I extract the GPU usage information from this field. For example, when gres/gpu = 1, it means that the job uses one GPU.
- A job is considered a GPU job if it need at least one GPU.

Analysis for requested resource: Required CPUs

Statistic	Requested CPUs
Observations	6,025,326
Mean	8.41
Std. Dev.	18.31
Min	0
Median (50%)	4
90th Percentile	16
95th Percentile	28
99th Percentile	80
Max	8,192

Table 5: CPU request summary

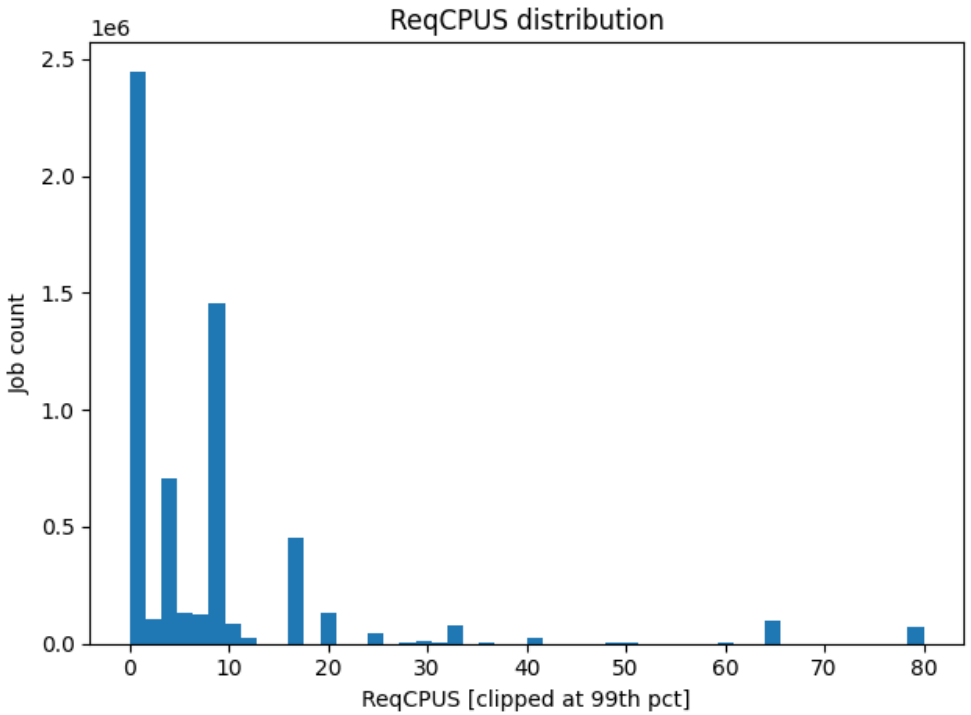

Figure 5: Distribution of requested CPUs

Analysis for requested resource: Required Nodes

Statistic	Requested Nodes
Observations	6,025,326
Mean	1.01
Std. Dev.	0.18
Min	0
Median (50%)	1
90th Percentile	1
95th Percentile	1
99th Percentile	2
Max	35

Table 6: Requested nodes summary

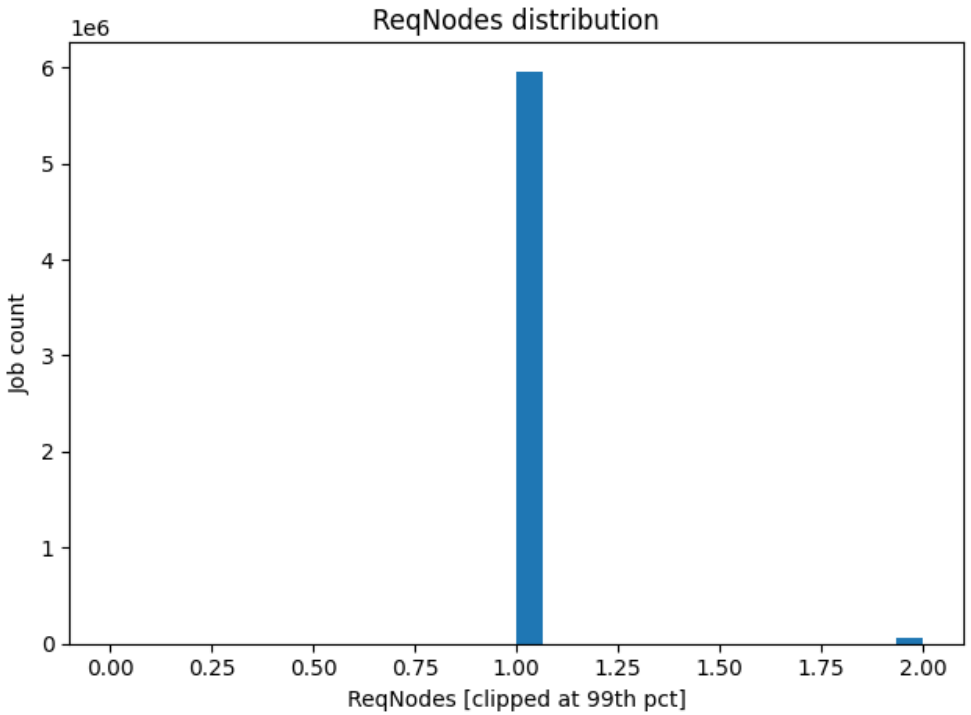

Figure 6: ReqNodes distribution

Analysis for requested resource: Required Memory (in GB)

Statistic	Requested Memory (GB)
Observations	6,025,325
Mean	52.97
Std. Dev.	318.84
Min	0.001
Median (50%)	16
90th Percentile	125
95th Percentile	180.44
99th Percentile	453.42
Max	555,550

Table 7: Requested memory summary

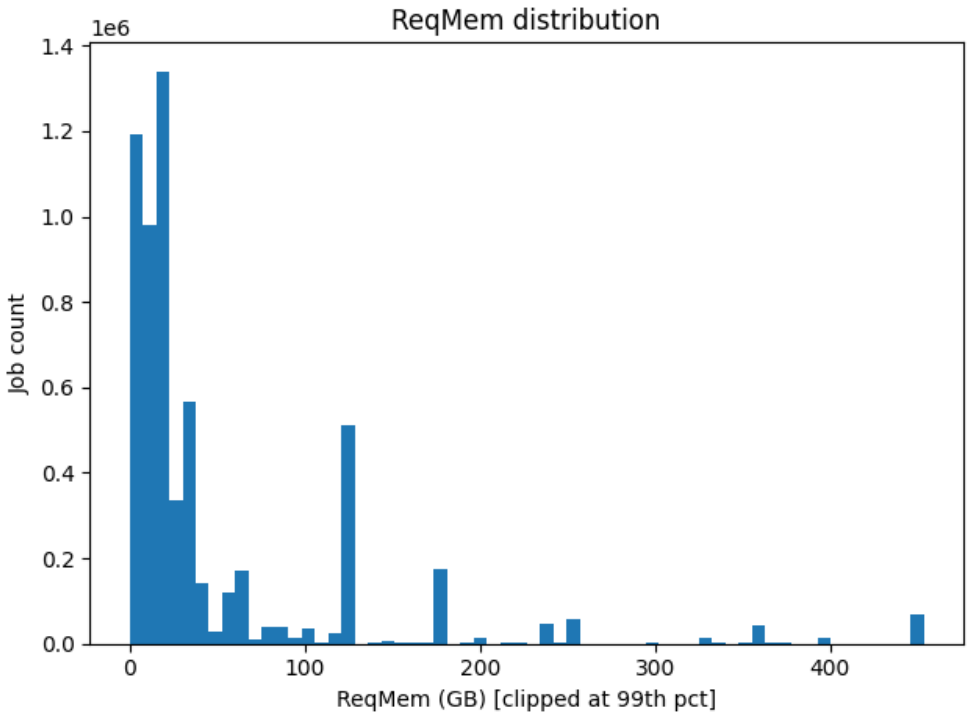

Figure 7: ReqMem distribution

Analysis for requested resource: Required GPUs

Requested GPUs	Number of Jobs
1	342,443
2	6,569
3	382
4	9,999
5	39
6	913
7	25
8	21,160
10	31
16	352
24	1,231
28	12
30	5
32	1
40	2
64	1

Table 8: GPU request distribution (GPU jobs only)

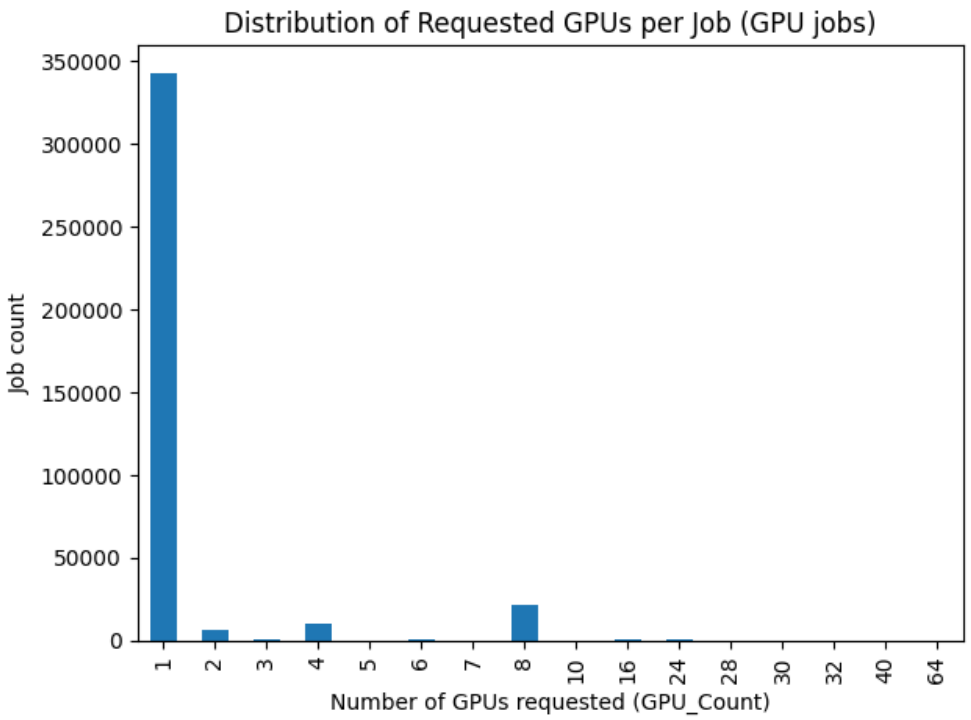

Figure 8: Distribution of Requested GPUs per Job

Wait time distribution for GPU jobs (in hours)

Statistic	Wait Time (hours)
Observations	351,136
Mean	8.81
Std. Dev.	42.80
Min	0.00
Median (50%)	0.087
90th Percentile	24.13
95th Percentile	48.57
99th Percentile	112.73
Max	6865.70

Table 9: GPU job wait time summary

Figure 9: Wait Time Distribution for GPU Jobs

Is there a significant difference in the waiting time between tasks that use only the CPU and those that require a GPU?

Before we start

- The dataset contains 383,167 GPU jobs and 5,642,159 CPU-only jobs. After cleaning invalid values, the analysis included 351,138 GPU jobs and 5,166,144 CPU-only jobs with valid waiting times.
- Jobs with GPU_Count > 0 were classified as GPU jobs, while jobs with GPU_Count $= 0$ were classified as CPU-only jobs.
- Because the waiting time data are non-normal and contain extreme values, Mann–Whitney U test was applied to determine whether the distributions of waiting times for CPU-only jobs and GPU jobs differ significantly.

Wait time comparison: GPU vs CPU jobs

Statistic	GPU Jobs	CPU-only Jobs
Observations	351,138	5,166,144
Mean	8.81	6.83
Std. Dev.	42.80	30.32
Median (50%)	0.087	0.44
90th Percentile	24.13	12.51
95th Percentile	48.57	29.12
99th Percentile	112.73	116.53
Max	6865.70	1541.16

Table 10: Comparison of wait time between GPU and CPU jobs

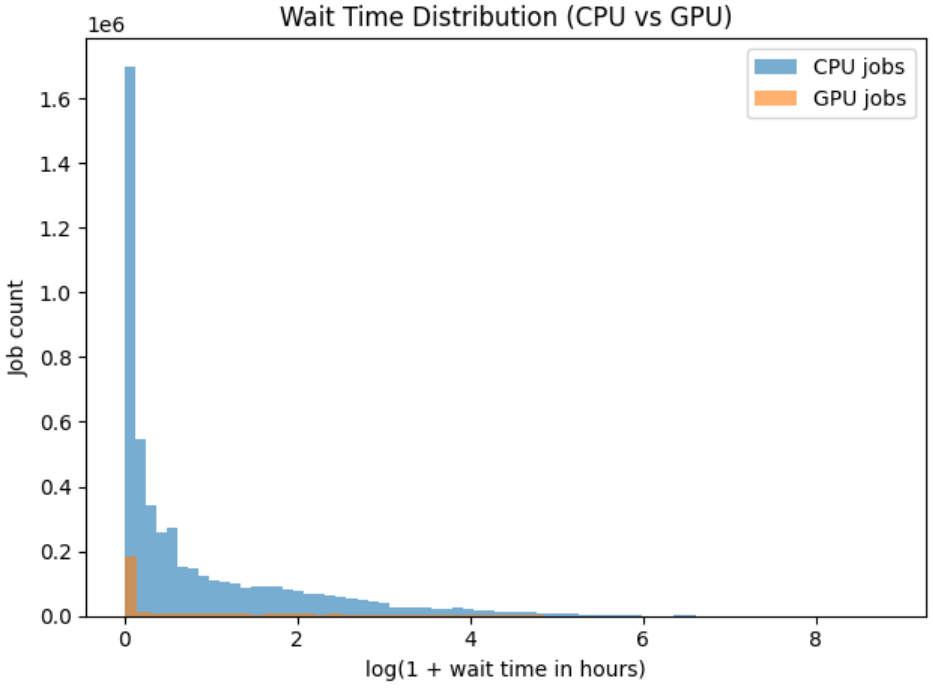

Figure 10: Wait Time Distribution: CPU vs GPU (after log(1+time))

Frame Title

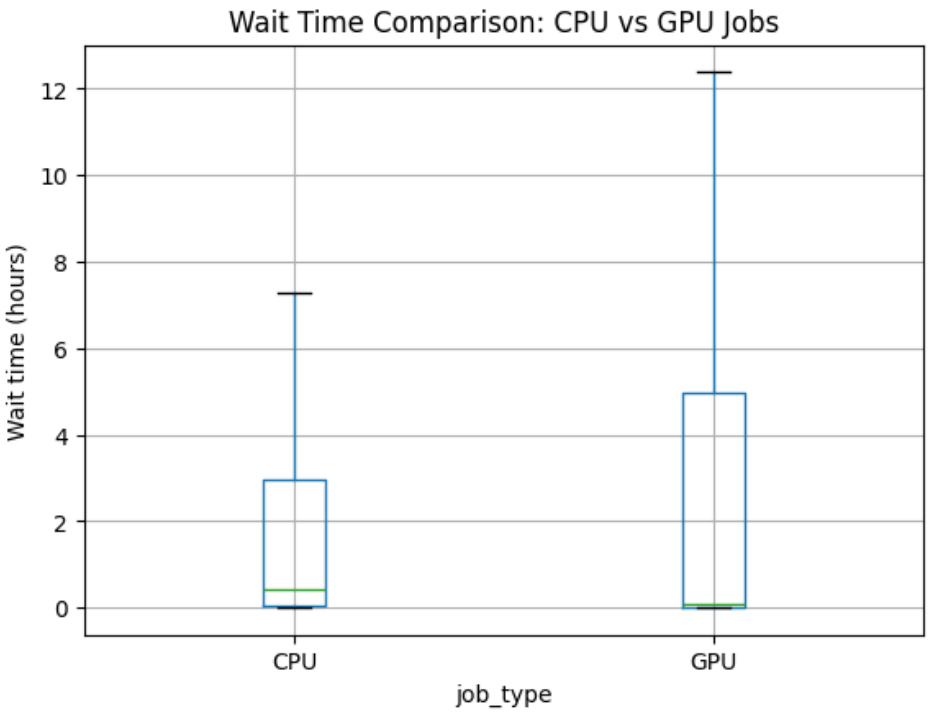

Figure 11: Boxplot comparison

Result

- The test returned a p-value of approximately 0, which is far below the conventional significance level of 0.05. Therefore, we reject the null hypothesis that the waiting time distributions for CPU and GPU jobs are the same.
- Overall, the results indicate that there is a statistically significant difference in waiting times between CPU-only jobs and GPU jobs. This difference is likely due to the higher demand and limited availability of GPU resources in the cluster environment, which can lead to longer queue times for GPU-based workloads.

Plan for future work

Plan for future work

1. Based on the available data, we can train a clustering model to predict job waiting time. For example, given job features (e.g., required GPUs, CPUs, memory, and nodes), our model can provide an estimated waiting time.
2. Providing recommendations for resource allocation and offer guidance for users, such as reference waiting times after submitting jobs.

You can find my code and slides on my github repo:
`https://github.com/Jiachen-Cong-UIUC/STAT427-Jiachen-Cong.git`

Thank You

Thank you for your attention!

Questions?